How can carbon quota pledge financing contribute to the two-stage production and carbon abatement of emission-dependent remanufacturing firms?

Abstract:

Due to the severe environmental pollution problem and the guidance of carbon emission reduction (CER) policies by the government, firms conduct carbon abatement investment (CAI) through different CER paths to realize low carbon production, in which capital constraint may be a significant factor that cannot be ignored. Thus, to rational use the carbon quota allocated, carbon quota pledge (CQP) acts as an effective financing method in supporting CERs. In this paper, based on carbon cap-and-trade mechanism, we establish two two-period manufacturing/ remanufacturing production decision models to contrastively evaluate the effectiveness of capital constraint and carbon quota pledge financing (CQPF) on the optimal production, CER and profits of firms under two CAI methods consisting of carbon emission purification and R&D investment. Our results indicate that CQPF more effectively enhance CERs and increase profits of firms when the self-owned capital is shortage, whereas it contributes to the increase in carbon emissions simultaneously. We also find that R&D investment has a cumulative effect, namely, the CER effect of the second period is more obvious than that of the first period, and the CERs and expected profits may even be higher than the case of carbon emission purification.

***Key words*:** Manufacturing/remanufacturing; Carbon emission purification; R&D investment abatement; Capital constrained; Carbon quota pledge financing

1. Introduction

Faced with increasingly stringent environmental protection requirements and severe public pressure to go green, General Secretary Xi Jinping proposed the goal to strive for the peak of carbon emissions by 2030 and strive for achieving carbon neutrality by 2060 at the 75th session of the United Nations General Assembly. In order to accelerate the realization of this goal, the Chinese government has enacted various carbon emission regulations to constrain firms' carbon emissions, and emphasized the need to give more play to the role of market mechanism in realizing the CER goal. In this situation, firms can offset the excess carbon emissions generated in production and operation activities by purchasing carbon emission quotas, or they can also reduce the carbon emissions by purchasing purification equipment of CERs or investing in R&D of low-carbon technologies, so as to achieving low-carbon green development. That is, to analyze the CER path of firms from the micro production level, CAI can be divided into three channels: R&D investment, purification equipment investment and behavioral abatement investment. Thus, due to the government's macro control and firms' micro CER measures, the low-carbon transformation of manufacturing firms is imperative. Among, through remanufacturing, firms can alleviate the aggravation of environmental burden caused by a large number of scrapped products and promote the recycling and echelon utilization of waste products (Van and Salomon, 1997; Chang et al., 2015). Thus, remanufacturing is an important development direction of low-carbon economy.

However, with the depth of the development of low-carbon system, the pressure on firms to reduce carbon emissions increases, capital constraint becomes a factor that cannot be ignored. The capital shortage of firms not only affect the output of products, but also may lead to the waste of other resources. Therefore, many firms attempt to solve or alleviate this issue through various financing methods (Xu and Fang, 2020; Xia et al., 2023). Nevertheless, most of them ignore the real role of carbon finance and do not rational use of carbon quota allocated. In practice, the firm may consider the real situation to solve the problem of capital shortage and study the CER strategies. The pledge

financing of carbon quota (hereinafter referred to as CQPF) is an innovative financing method in carbon market that takes the obtained carbon quota as collateral, and it can flexibly set the repayment period according to the emission reduction output (Cao and Yu, 2019). Therefore, CQP can effectively help firms to revitalize carbon quota assets and solve the problem of financing difficulties for small and medium-sized firms. Meanwhile, in terms of banks, the carbon quota obtained by firms belongs to the existing assets, which is easy to supervise in the pledge process and has little risk of realization. Therefore, this financing mode has attracted more and more attention in recent years.

As for manufacturing/remanufacturing production decision problems in a low-carbon environment, most of the scholars focus on the integration of carbon emission constraints and production operations. Also, there are some studies exploring the impact of capital constraints on firms' production decisions, whereas few research consider the impact of different CER methods on the two-stage manufacturing/remanufacturing production decisions. Furthermore, fewer research probe into the benefits of carbon finance for firms' production and carbon abatement (PCA). Therefore, this paper considers two methods of CAI consisting of carbon emissions purification and R&D investment, exploring the integrated optimization of manufacturing/remanufacturing and CER in the case of carbon pledge financing, which aims to achieve the goal of improving firms profits and reducing carbon emissions in the process of manufacturing/remanufacturing operation by using carbon finance. The result has very important theoretical significance and practical value.

2. Literature review

Two main streams of researched are related to our work, including the researches exploring the CAI on production under carbon emission regulation, and the researches addressing the supply chain financing with considering CER strategies. We next review each stream and position our paper accordingly.

2.1. CAI on production

In the former stream of research, CER decisions are explored in many studies. Most of the literature related to CER decisions on the firm level mainly focus on the carbon emissions policies under different carbon regulations (Chen et al., 2021; Fan et al., 2023; Zhang et al., 2024; He et al., 2024; Zhang et al., 2024; Yang and Xu, 2024).

Another group of scholars focus on CER path. That is, CAI can be divided into three channels: research and development (R&D) investment, purification equipment investment and behavioral abatement investment. Among, fewer scholars investigate the purification of the carbon emission from the perspective of "end treatment" and examine the contribution of end-of-pipe abatement to the reduction of pollutant emissions under different environmental regulations (Turken et al., 2020; Wang et al., 2021). Similar, Zeng et al. (2020) emphasize the crucial role of end-of-pipe abatement technologies and investigate the effect of emission taxes and standards on the adoption of end-of-pipe abatement technology when the new emissions abatement technology are uncertain. With regard to the research of R&D investment, Qiao et al. (2021) examine sustainability coordination among three major supply chain parties where both the supplier and the manufacturer embark on green investment to reduce carbon emissions; Pan et al. (2021) reveal the impact of carbon dioxide reduction constraint on enterprises' R&D investment; By constructing an evolutionary game model, Jiao et al. (2023) study the behavior choice of carbon reduction of waste power battery recycling enterprises from internal R&D motivation and external regulation perspectives; Cheng and Wang (2023) emphasize that for socially

responsible enterprises, green technology and trade-in programs are two ways to benefit the environment. Similarly, the research of emission-based R&D investment are also examined by Li et al. (2021), Ma et al. (2021) and Jiang et al. (2023). Though PCA strategies are analyzed form different perspectives in the above researches, as an effective way to solve the issue of capital shortage and make full use of carbon quota allocated, CQP is not considered.

2.2. Supply chain financing with CER strategy

Up to now, there are quite a few literature on firms' PCA decisions under capital constraints. For example, to meet consumers' low-carbon preferences, Dong et al. (2023) consider a supply chain conducting low-carbon investments and solve their financing constraints by utilizing portfolio financing; Similarly, Zhang et al. (2024) explore the manufacturer' equilibrium emission-reduction investment strategies and endeavor to solve the issue of the retailer' capital constraint by three scenarios; Gao et al. (2022) investigate a new financing service- online peer to peer lending- for SEMs and investigate the effects in funding low-carbon supply chains through multi-level game models. Besides, Tang et al. (2023), Zou et al. (2021), Shi et al. (2023) and Qin et al. (2020) are among the researches doing similar studies.

Though supply chain financing with low carbon policies are modeled from different aspects, above studies from strategic perspective did not pay enough attention on carbon finance, which will greatly promote the low carbon economy. No doubt that carbon tax policy can't fully satisfy the demand of low-carbon economic transformation, the support of financial policies such as banks and monetary policy are also essential (Campiglio, 2016). In this way, carbon finance plays a significant role in motivating upstream and downstream enterprises in a capital-constrained supply chain to reduce carbon emissions for sustainable development (Wang et al., 2024a, b). In terms of construction machinery remanufacturing firms, Fu et al. (2023) examine the impacts of different cooperation financing and carbon emission reduction strategies on the operational decisions in a closed-loop supply chain in which carbon fund is employed; Wang et al. (2024) explore two carbon quota allocation methods of a capital-constrained e-commerce supply chain that obtains financing through carbon rights buyback by game models.

Among them, notice that the carbon quota as a pledge is considered in a supply chain to support the manufacturer emission reduction as well as responsible consumption and production, which include bank carbon financing mode and a supply chain carbon financing mode (Shang et al., 2024). Likewise, Fu et al. (2023) suppose the emission-dependent remanufacturer adopts the pure carbon asset pledge financing to solve capital shortage problem, and explore the optimal production and CER decisions of the engineering machinery remanufacturer. Similar research has been done by Xu et al. (2024). Different from above studies, in this paper, the two-period manufacturing/remanufacturing PCA problems are considered simultaneously and different methods of CAI are compared with considering CQPF.

2.3. Discussion

The literature above contribute to the studies of CER and financing in a capital-constrained supply chain from different aspects. Most of them, however, ignore the real role of carbon finance and do not rational make use of carbon quota allocated. Firstly, most of the literature related to CER decisions on the firm level mainly focus on the carbon emissions policies, little has focused on specific CER paths from different perspectives, even less consider the issue of pledging allocated carbon quota. Secondly,

although quite a number of studies have explored the PCA decisions under capital constraints of firms, comparing and analyzing the advantages and disadvantages of different financing methods, few have focus on the practical financing tool-carbon finance, which can provide financial support for CERs of emission-based manufacturing firms in china. In the field of carbon finance, most of the literature focus on carbon rights buyback, carbon quota offsetting and carbon trade finance, only a few papers consider the pledge financing of carbon quota.

To the best of our knowledge, our research innovatively jointly consider CAI and CQPF in a two-period manufacturing/remanufacturing production system. Besides, we compare the impacts of two CAIs on the production decision to develop insights for flexible use of the emission quota allocated by the government. That is, our findings not only help the firm make better CER and CQPF strategies in environmental protection, but also provide useful guidelines for the government to adjust the carbon price to enhance the benefit of firms both economically and environmentally.

3. Notation and Problem description

We consider a monopolistic manufacturer in a hybrid manufacturing/remanufacturing system in which the firm produces new products in the first period, collects a certain ratio of used products for remanufacturing from the first period and produces news products simultaneously in the second period. Assume that the new and remanufactured products are sold in the same market over two periods. Since the valuations and features of the new and remanufactured products are different, so the two competing products, influence to each other. In addition, during the remanufacturing process, losses are incurred due to either process handling problems or the unexpected problems, such as the inferior quality of returns. Thus, the production quantity of remanufactured products in the second period are usually less than the returns ($q_{2r} \leq \sigma q_{1n}$). Specially, we don't consider the risk of overproduction or out-of-stock for the remanufacturer and the stock costs of raw materials and returns.

At the beginning of each period, the government allocates a certain carbon allowances free of charge (E_g). Thus, the manufacturer is constraint by a carbon cap where the carbon caps over the two periods are the same. Driven by this situation, the manufacturer may reduce carbon emissions by making R&D investment or carbon emission purification in order to respond to the government's call for energy conservation and CER and optimize the operation cost of the firm.

In the first period, the firm owns ample self-owned capital (B_1) for PCA. Thus, the initial capital of second period (B_2) derived from partial expected profit (retained earnings) and the remaining initial capital of the first period. In this case, if the initial capital in period 2 is insufficient for abatement investment, the manufacturer finances by pledging a certain ratio of carbon quota at the beginning of period 2. In other word, the firm signs a pledge contract with its financial institution, and the carbon credits pledged will arrive in an instant. The quantity of carbon pledged (E_p) may be the carbon cap allocated by the government plus the margin of carbon quota in period 1. It is noted that carbon credits pledged can only be used to reduce emissions, and the rest of the cost will be undertaken by the self-owned capital. In the sales period, assume that the firm can receive payment in time and he would conduct carbon trading after repaying loans to the financial institution and redeeming the quantity of carbon pledged. So far, the entire amount of principal and interest should be $(1+r)c_m\beta E_f$ at the end of the pledge. In this pledge business, the financial institution determines the mortgage rate β and the interest rate r, and firm carries out carbon emission reducing, both parties are risk neutral and complete information symmetry.

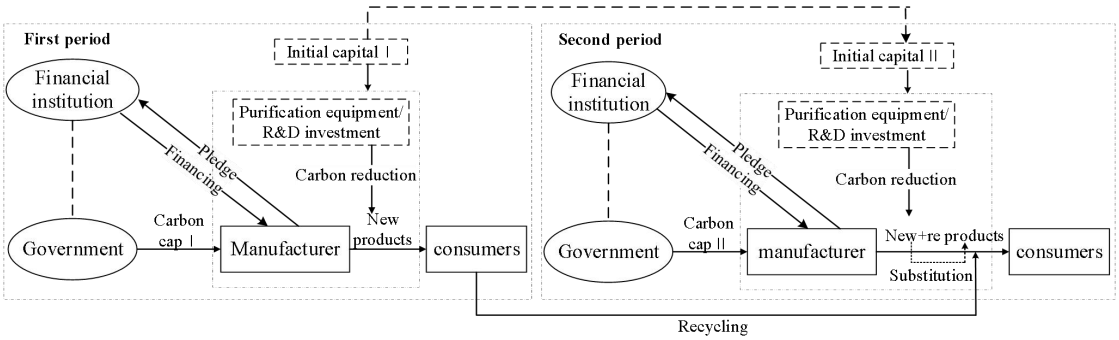

Fig. 1. The whole process of production operation and carbon quota pledge

It is noted that the manufacturer conducts R&D investment by making a one-time investment in new technology, equipment or machinery (Li et al., 2021). Without loss of generality, the investment $\frac{1}{2}n_i\tau_i^2$ are related to unit mitigation rate (τ_i) and the CERs depend on investment. Differ from the R&D investment, the investment of purification (c_pE_{ip}) are concerned with carbon emissions from purification (E_{ip}) and unit purification cost (c_p). Furthermore, based on the cap-and-trade regulation, the manufacturer may purchase quota from the carbon trading market if the total carbon emissions surpass the original allowance, or he may exchange carbon emissions for the value of monetary after carbon reducing in the second period. Assume carbon trading price being an exogenous variable and is determined by the carbon market. In short, both the efficiency of CER/the level of purification and the unit carbon trading price have an impact on its CER effect. The whole process of manufacturing/remanufacturing operation and CQP is shown in Fig. 1.

The notations in the modeling setup are summarized in Table 1 below. Some other notations used in this paper are defined additionally when using them.

Table 1
Notation.

Indices	
i	Index for production period, $i=1, 2$ for the 1st and 2st production period respectively.
j	Index for ways of CER, $j=(1), (2)$ for the carbon emissions purification and R&D investment abatement respectively.
k	$k=a, b$ respect the case of ample funded and underfunded respectively.
Parameters	
c_n / c_r	Unit production cost of new/remanufactured products, $c_r = c_n - \Delta$, Δ denotes the cost saving per unit remanufactured product, suppose $\Delta > 0$.
p_{in}^{jk} / p_{ir}^{jk}	Unit sales price of new/remanufactured products for period i, mode j and case k.
Q_i	Size of potential market size for period i.
σ	Recovery rate of new products for period 1, $0<\sigma<1$
ε	Recovery cost coefficient of new products for period 1, $\varepsilon>0$.
k	Consumer acceptance of remanufactured products, $0<k<1$
e_n / e_r	Unit carbon emissions of new/remanufactured product, $e_r=\tau e_n$, τ denotes the remanufacturing

	carbon intensity, suppose $0<\tau<1$.
E_g	Free carbon cap allocated to the manufacturer for each period.
c_{im}	Unit carbon trading price and unit price of CQP for period *i*.
c_p	Unit carbon emission purification cost
α_i	Carbon emission purification level for period *i*, $0 < \alpha_i < 1$
n_i	Cost coefficient of R&D investment abatement for period *i*, $n_i>1$.
E_{ip}	carbon emission quota of purification for period *i*.
E_{if}^j	The quantity of CQP for period *i*, mode *j*.
β	Pledge rate of carbon quota, suppose the pledge rate β is constant. $0<\beta<1$
B_i^j	Initial capital by the manufacturer for period *i*, mode *j*.
ϕ	The retained earnings ratio of the first period of the manufacturer, $0\le\phi\le1$
r	Financing rate of CQP, $0<r<1$
T_p^j	The pledge period for mode *j*. The pledge period is the production period (T_{21}) plus the sales period (T_{21}).
π_i^{jk}	The expected profit of manufacturer for period *i*, mode *j* and case *k*.
Decision variables	
q_{in}^{jk} / q_{ir}^{jk}	Production quantity of new/remanufactured products for period *i*, mode *j* and case *k*.
τ_i^k	Unit mitigation rate for period *i* and case *k*, $0 < \tau_i^k < 1$.

4. Establishing and solving the model

According to problem description and assumption, we explore the manufacturer's jointly PCA decisions over the two periods by considering two different CAIs. Since the initial capital of the second period relies on the first period. Thus, the manufacturer makes the production decision beginning from the first period.

4.1. Carbon emissions purification

In this subsection, we consider the situation of carbon emission purification by the remanufacturing firm. Assume that the level of carbon emission purification α over the two periods are the same.

4.1.1. The first period

In the first period, the self-owned capital of the firm B_1 is ample and only the new products are produced. Following Dou et al. (2019), the price function of the new products is assumed as $p_{1n}^{(1)} = 1 - q_{1n}^{(1)}$. The profit-maximizing problem of the remanufacturing firm in this case can be given by:

$$max\,\Pi(q_{1n}^{(1)}) = \underbrace{(p_{1n} - c_n)q_{1n}}_{1} - \underbrace{c_p E_{1p}}_{2} - \underbrace{c_m[(1-\alpha)e_n q_{1n} - E_g]}_{3} \quad (1)$$

$$s.t.\quad c_n q_{1n} - c_p E_{1p} < B_1$$

In the objective function, the first term represents the profits from selling new products in the first period, the second term is the cost of carbon emission purification in this period, and the third term is cost or profit associated with carbon trading in period 1. Constraint implies that the self-owned capital

in the first period are well-funded for PCA. For simplicity, we assume the carbon emissions purification equals to the carbon emissions in both the first and the second period, which represents $E_{ip} = E_i(q)$, $i=1,2$.

The optimality of objective function (1) can be characterized by the Karush-Kuhn-Tucker (KKT) conditions, and this optimization leads to the following Theorem. The proof of Theorem 1, as well as those for other theorems, can be found in the Appendix A.

Theorem 1. When the remanufacturing firm produces only new products and conduct carbon emission purification in the first period, there exists a unique optimal new production $q_{1n}^{(1)}=\frac{1}{2}(\psi_2-\psi_1)$ depending on the carbon trading price c_m. Specially, the optimal PCA decisions can be defined as strategy PI.

Theorem 1 indicates that to ensure the optimal new production is positive, it needs to satisfy the condition $c_m<c_{m1}$. In which, $cm_1=\frac{\psi_2}{e_n(1-\alpha)}$, $\psi_1=c_m e_n(1-\alpha)$, $\psi_2=Q-c_p e_n-c_n$.

Thus, the optimal total CERs and profit can be given as:

$$E_{1a}^{(1)}=\frac{1}{2}\alpha e_n[\psi_2-\psi_1],\quad \pi_1^{(1)}=\frac{1}{2}(\psi_1-\psi_2)^2+c_m E_g$$

The remaining capital of the first period is $B_1^{(1)'}=B_1-c_n q_{1n}-c_p E_{1p}$, thus the initial capital of the second period is derived from the partial profits and the remaining initial capital of period 1, that is $B_2^{(1)}=\phi\Pi_1+B_1^{(1)'}$.

4.1.2. The second period

In the second period, the remanufacturing firm not only produces new products, but also collect returns derived from the first period to produce remanufactured products. The price functions of the new and remanufactured products are assumed as $p_{2n}=Q-q_{2n}-kq_{2r}$ and $p_{2r}=k(Q-q_{2n}-q_{2r})$, respectively. Similar assumptions are widely adopted in the previous researches (Dou et al., 2019 and Chang et al., 2017).

(1) Sufficient capital

In this case, the initial capital of the second period is not less than the production and carbon purification abatement. Thus, the mathematical model for maximizing the total profit in this case is given as follows:

$$max\,\Pi(q_{2n}^{(1)},q_{2r}^{(1)})=\underbrace{(p_{2n}-c_n)q_{2n}+(p_{2r}-c_r)q_{2r}}_{1}-\underbrace{\varepsilon\sigma^2}_{2}-\underbrace{c_p E_{2p}}_{3}-\underbrace{c_m[(1-\alpha)(e_n q_{2n}+e_r q_{2r})-E_g]}_{4} \quad (2)$$

$$\text{s.t.}\quad \begin{cases} c_n q_{2n}+c_r q_{2r}+\varepsilon\sigma^2+c_p E_{2p}\le B_2 \\ \sigma q_{1n}=q_{2r} \\ q_{2n},q_{2r}\ge 0 \end{cases}$$

In the objective function, the first term is the profits generated from selling new and remanufactured products in the second period. The second term represents the collection and inspection cost of the returns. The third term is the cost of carbon emission purification in this period, and the last term is cost or profit associated with carbon trading in period 1. In the constraints, for the sake of brevity, we assume that $\sigma q_{1n}=q_{2r}$ where the remanufactured product quantity is equal to the collected

returns from period 1. Finally, the non-negativity constraints ensure the production quantities to be greater than or equal to zero. After solving model (2), we obtain the following Theorem:

Theorem 2. When the remanufacturing firm produces both new and remanufacturing products and conduct carbon emission purification in the second period, if the initial capital of this period is ample, the optimal production decisions are shown in Table 2, which depend on carbon price c_m and rely partly on the remanufacturing carbon intensity τ. The critical of them define two decision strategies, namely PIIA1 strategy and PIIA2 strategy.

Theorem 2 offers a menu of optimal PCA decisions. The first column of Table 2 shows the optimal production strategies in the second period, and the second column displays the optimal production quantities under corresponding strategy. In which, $\psi_3=(\Delta-c_n)q_{1n}\sigma$, $\psi_4=c_p e_n+c_n$, $\psi_5=-\varepsilon\sigma^2+\phi\pi_1+B_1$.

Associated with Theorem 1, we derive the Proposition 1 as follows:

Table 2

The optimal production and carbon abatement decisions under carbon purification.

Carbon purification		The optimal strategy in period 2	The optimal production quantities in period 2
Period 2	Sufficient capital	PIIA1	$q_{2n}^{(1)a_1}=\frac{\psi_3-c_p e_n\tau q_{1n}\sigma-\psi_4 q_{1n}+\psi_5}{\psi_4}$, $q_{2r}^{(1)a_1}=\sigma q_{1n}^{(1)}=\frac{1}{2}\sigma(\psi_2-\psi_1)$
		PIIA2	$q_{2n}^{(1)a_2}=\frac{1}{2}(\psi_2-\psi_1)-k\sigma q_{1n}$, $q_{2r}^{(1)a_2}=q_{2r}^{(1)a_1}$
	Insufficient capital	PIIS1	$q_{2n}^{(1)b_1}=\frac{\psi_3-c_p e_n\tau q_{1n}\sigma-\psi_4 q_{1n}+\psi_5+E_g\beta c_m}{\psi_4}$, $q_{2r}^{(1)b_1}=\frac{1}{2}\sigma(\psi_2-\psi_1)$
		PIIS2	$q_{2n}^{(1)b_2}=\frac{\psi_3-\psi_4 q_{1n}+\psi_5}{c_n}$, $q_{2r}^{(1)b_1}=\frac{1}{2}\sigma(\psi_2-\psi_1)$
		PIIS3	$q_{2n}^{(1)b_3}=\frac{1}{2}[\psi_2-\psi_1-k\sigma q_{1n}]$, $q_{2r}^{(1)b_1}=\frac{1}{2}\sigma(\psi_2-\psi_1)$

Proposition 1. Under the PII strategy with the case of sufficient capital, the manufacturer's optimal production, CERs and capital utilization policies are as follows:

(1) If $c_m<c_{m1}$

(i) when $\tau 2<\tau<\tau 1$, the manufacturer should offer the PIIA2 strategy that the initial capital of second period is equal to the production and carbon purification abatement.

(ii) otherwise, the manufacturer should offer the PIIA1 strategy that the initial capital of second period is abundant for production and carbon purification abatement.

The critical values of c_m and τ defining two decision strategies are shown in Table 3, in which let $\tau 1=\frac{\psi_3-\psi_4 q_{1n}+\psi_5}{c_p e_n q_{1n}\sigma}$, $\tau 2=\frac{\psi_4[2kq_{1n}\sigma+(1-\alpha)c_m e_n-\psi_2-2q_{1n}])+2\psi_3+2\psi_5}{2c_p e_n q_{1n}\sigma}$.

Proposition 1 characterizes the optimal strategies for the situation of ample capital in the second period by matching carbon price and the remanufacturing carbon intensity. Table 3 indicates that the parameters of c_m and τ affect the optimal decisions in order to ensure the positive values of decision

variables. When the remanufacturing carbon intensity is constraint by a certain ratio, the funds utilizing are just be the initial capital in period 2, otherwise, the use of capital is independent of the initial capital in the second period. It is noted that all the above optimal decisions come true if carbon pricing $c_m<c_{m1}$.

Thus, the optimal total CERs can be given as:

$$E_{2a}^{(1)a_1} = \alpha e_n[\frac{\psi_3 - c_p e_n \tau q_{1n}\sigma - \psi_4 q_{1n} + \psi_5}{\psi_4} + \frac{\psi_2 - \psi_1}{2\tau\sigma}], \quad E_{2a}^{(1)a_2} = \frac{1}{2}\alpha e_n[(Q-\psi_4)(e_n\sigma\tau+1)-\psi_1(\sigma\tau+1)-2kq_{1n}\sigma]$$

We will next analyze the situation of insufficient capital and CQP in the second period.

Table 3

The impact of c_m and τ on optimal strategies under carbon purification.

c_m	τ	The optimal strategies in period 2
$c_m < c_{m1} = \frac{\psi_2}{e_n(1-\alpha)}$	$\tau 2<\tau<\tau 1$	PIIA1
	—	PIIA2
$c_m < c_{m1}$	$\tau 4<\tau<\tau 3$	PIIS1
	—	PIIS2
$c_m < c_{m2} = \frac{1-\psi_4 - 2kq_{1n}\sigma - 2q_{2n}^{(2)b_2}}{e_n(1-\alpha)}$	—	PIIS3

(2) Insufficient capital and CQP

In this case, we consider the initial capital of second period insufficient, the remanufacturing firm pledge a certain ratio of carbon quota for carbon emission purification. It is important to notice that the price of CQP is the same as the carbon trading price for each unit of carbon quota in this period. Thus, the mathematical model for maximizing the total profit in this case is given as follows:

$$max\,\Pi(q_{2n}^{(1)}, q_{2r}^{(1)}) = (p_{2n} - c_n)q_{2n} + (p_{2r} - c_r)q_{2r} - \varepsilon\sigma^2 - c_p E_{2p} - c_m[(1-\alpha)(e_n q_{2n} + e_r q_{2r}) - E_g] - rc_m\beta E_f \quad (3)$$

$$\text{s.t.} \begin{cases} c_n q_{2n} + c_r q_{2r} + \varepsilon\sigma^2 \le B_2 \\ \sigma q_{1n} = q_{2r} \\ c_n q_{2n} + c_r q_{2r} + \xi\sigma^2 + c_p E_{2p} = B_2 + c_m\beta E_f \\ 0 \le E_f \le E_g \\ q_{2n}, q_{2r} \ge 0 \end{cases}$$

In the objective function, the main difference between model (3) and model (2) is the extra term of CQP interest $rc_m\beta E_f$. In the constraints, the initial capital of the second period plus carbon credits pledged are just enough to afford the production and carbon purification abatement in period 2. $E_f \le E_g$ implies that the quantity of carbon pledged is less than or equal to the carbon cap in period 2.

The solution method for solving model (3) is similar to that for solving model (2), so proofs for model (3) are omitted for brevity and are available from authors. Thus, we obtain the following Theorem:

Theorem 3. When the manufacturer produces both new and remanufacturing products and conduct carbon emission purification in the second period, if the initial capital of this period is insufficient, the optimal production decisions are shown in Table 2, which are significantly affected by carbon price c_m and the remanufacturing carbon intensity τ. The critical of them define three decision strategies,

namely PIIS1 strategy, PIIS2 strategy and PIIS3 strategy.

Theorem 3 offers three optimal strategies for the case of CQP in period 2. Associated with Theorem 3, we derive the Proposition 2 as follows:

Proposition 2. Under the PII strategy with the case of insufficient capital, the manufacturer's optimal production, CERs and capital utilization policies are as follows:

(1) If $c_m<c_{m1}$

(i) when $\tau 4< \tau< \tau 3$, the manufacturer should offer the PIIS1 strategy that the initial capital of second period is ample for production and collection and the quantity of carbon pledged is equal to the carbon cap.

(ii) otherwise, the manufacturer should offer the PIIS3 strategy that the initial capital of second period is abundant, whereas the quantity of carbon pledged is less than the carbon cap.

(2) If $c_m<c_{m2}$, the manufacturer should offer the PIIS2 strategy that the initial capital of second period just meet production and collection and the quantity of carbon pledged is less than the carbon cap.

The critical values of c_m and τ defining the three decision strategies are shown in Table 3.

Proposition 2 shows the optimal strategies for the situation of insufficient capital and carbon emission pledge in the second period. Table 3 also characterize the strategy PIIS3, strategy PIIS3 and strategy PIIS3 by matching the parameters of c_m and τ to ensure the positive values of decision variables. If $c_m<c_{m1}$, when the remanufacturing carbon intensity is constraint by certain values, the initial capital in period 2 is ample for production and collection and we pledge the overall carbon quota for carbon emission purification, otherwise, we only pledge a certain ratio of carbon quota. If $c_m<c_{m2}<c_{m1}$, that is the carbon trading price is lesser, the initial capital is equal to production in the second period and we only pledge a certain ratio of carbon quota for carbon emission purification.

Thus, the optimal total CERs and the quantity of carbon pledged can be given as:

$$E_{2a}^{(1)b_1} = \frac{\alpha e_n(\psi_3 + c_n\tau q_{1n}\sigma - \psi_4 q_{1n} + \beta c_m E_g + \psi_5)}{\psi_4}, \quad E_{2a}^{(1)b_2} = \frac{\alpha e_n(\psi_3 + c_n\tau q_{1n}\sigma - \psi_4 q_{1n} + \psi_5)}{c_n}$$

$$E_{2a}^{(1)b_3} = \frac{1}{2}\alpha e_n(\psi_2 - \psi_1 + \omega_4)$$

$$E_f^{(1)b_1} = E_g, \quad E_f^{(1)b_2} = \frac{c_p e_n[\psi_3 + (c_n\tau - \psi_4)q_{1n}\sigma + \psi_5]}{c_m c_n \beta}$$

$$E_f^{(1)b_3} = \frac{\frac{1}{2}\psi_4(\psi_2 - \psi_1) + (\frac{1}{2}\psi_4\omega_4 - \psi_3 - c_n\tau q_{1n}\sigma) + \psi_4 q_{1n} - \psi_5}{c_m\beta}$$

We will next analyze another method of carbon emission abatement--R&D investment abatement.

4.2. R&D investment abatement

In this subsection, we examine the situation of R&D investment abatement by the remanufacturing firm. In this scenario, R&D investment abatement is a process that requires long-term investment accumulation. Following Li et al. (2021) and Chen et al. (2017), R&D investment of the second period may be a further expansion of the research in the first period, the CER effect of period 2 is better than that of period 1. Thus, the cost coefficient of CER in the second stage (n_2) will be lesser than that in the first period (n_1) for the remanufacturing firm, that is $n_2<n_1$.

Based on problem description and assumption, this subsection makes the production decision

beginning from the first period, which is the same as 4.1.

4.2.1. The first period

The first period only produces new products, and the price function of the new products is assumed as $p_{1n}^{(2)} = 1 - q_{1n}^{(2)}$. The amount of R&D investment to reduce emissions in this period is defined as $\frac{1}{2}n_1\tau_1^2$. The profit-maximizing problem of the remanufacturing firm in this case can be given by:

$$max\,\Pi(q_{1n}^{(2)}) = (p_{1n} - c_n)q_{1n} - \frac{1}{2}n_1\tau_1^2 - c_m[(1-\tau_1)e_n q_{1n} - E_g] \quad (4)$$

$$s.t.\quad c_n q_{1n} + \frac{1}{2}n_1\tau_1^2 < B_1$$

The model construction and solution method for solving model (4) is similar to that for solving model (1) in situation 4.1. The optimality of objective function (4) characterized by the KKT conditions leads to the following Theorem.

Theorem 4. When the remanufacturing firm produces only new products and conduct R&D investment abatement in the first period, there exists a unique optimal new production $q_{1n}^{(2)} = \frac{n_1\omega_1}{\omega_2}$ and mitigation rate $\tau_1^{(2)} = \frac{c_m e_n \omega_1}{\omega_2}$ depending on the carbon trading price c_m. Specially, the optimal PCA decisions can be defined as strategy RI.

In Theorem 1, we assume that $2n_1 - c_m^2 e_n^2 > 0$, thus it needs to satisfy the conditions $c_m < c_{m3}$ and $c_m < c_{m4}$ to ensure $q_{1n}^{(2)} > 0$ and $\tau_1^{(2)} > 0$, In addition, in order to satisfy $\tau_1^{(2)} < 1$, it should meet the condition $c_m > c_{m5}$, in which, $c_{m3} = \frac{1-c_n}{e_n}$, $c_{m4} = \sqrt{\frac{2n_1}{e_n^2}}$, $c_{m5} = \frac{2n_1}{e_n(Q-c_n)}$ $\omega_1 = Q - c_n - c_m e_n$, $\omega_2 = 2n_1 - c_m^2 e_n^2$

Thus, the optimal total CERs and profit can be given as:

$E_{1a}^{(2)} = \frac{c_m e_n^2 n_1 \omega_1^2}{\omega_2^2}$, $\pi_1^{(2)} = \frac{2E_g c_m \omega_2 + n_1\omega_1^2}{2\omega_2}$

In this case, similar to strategy PI, the remaining capital of period 1 is $B_1^{(1)'} = B_1 - c_n q_{1n} - \frac{1}{2}n_1\tau_1^2$, thus the initial capital of the second period is $B_2^{(2)} = \phi\Pi_1 + B_1^{(1)'}$.

4.2.2. The second period

In the second period, the remanufacturing firm also produces new products and remanufactured products simultaneously and conduct R&D investment abatement.

(1) Sufficient capital

In this case, the initial capital of second period is not less than the production, collection and R&D investment abatement. Thus, the optimization model in this case can be formulated as:

$$max\,\Pi(q_{2n}^{(2)}, q_{2r}^{(2)}) = (p_{2n} - c_n)q_{2n} + (p_{2r} - c_r)q_{2r} - \varepsilon\sigma^2 - \frac{1}{2}n_1\tau_1^2 - c_m[(1-\tau_1)(e_n q_{2n} + e_r q_{2r}) - E_g] \quad (5)$$

$$\text{s.t.}\quad \begin{cases} c_n q_{2n} + c_r q_{2r} + \varepsilon\sigma^2 + \frac{1}{2} n_1 \tau_1^2 \le B_2 \\ \sigma q_{1n} = q_{2r} \\ q_{2n}, q_{2r} \ge 0 \end{cases}$$

The description of objective function and constraints have been detailed explained in the situation of carbon emission purification, here no longer etc. After solving model (5), we obtain the following Theorem:

Theorem 5. When the remanufacturing firm produces both new and remanufacturing products and conduct R&D investment abatement in the second period, if the initial capital of this period is ample, the optimal production decisions are shown in Table 4, which is defined as strategy RIIA1.

Theorem 5 offers a menu of optimal production and R&D investment abatement decisions for the case of sufficient capital in period 2. In Table 4, $\omega_3 = 2n_2 - c_m^2 e_n^2$, $\omega_4 = 2q_{1n}\sigma(\tau - k)$.

Table 4

The optimal production and carbon abatement decisions under R&D investment.

R&D investment		The optimal strategy in period 2	The optimal production quantities in period 2
Period 2	Sufficient capital	RIIA1	$q_{2n}^{(2)a} = \frac{n_2\omega_1 - (2kn_2 - c_m^2 e_n^2 \tau) q_{1n}\sigma}{\omega_3}$, $q_{2r}^{(2)a} = \frac{n_1\omega_1}{\omega_2}$, $\tau_2^{(2)a} = \frac{c_m e_n [\omega_1 + \omega_4]}{\omega_3}$
	Insufficient capital	RIIS1	$q_{2n}^{(2)b_2} = \frac{\psi_3 - \psi_4 q_{1n} + \psi_5}{c_n}$, $q_{2r}^{(2)a} = \frac{n_1\omega_1}{\omega_2}$, $\tau_2^{(2)b_2} = \frac{c_m e_n(\psi_3 + c_n \tau q_{1n}\sigma) - \psi_4 q_{1n} + \psi_5}{c_n n_2}$
		RIIS2	$q_{2n}^{(2)b_3} = \frac{n_2\omega_1 - (2kn_2 - c_m^2 e_n^2 \tau) q_{1n}\sigma}{\omega_3}$, $q_{2r}^{(2)a} = \frac{n_1\omega_1}{\omega_2}$, $\tau_2^{(2)b_3} = \frac{c_m e_n [\omega_1 + \omega_4]}{\omega_3}$

Associated with Theorem 5, we derive the Proposition 3 as follows:

Proposition 3. Under the RIIA1 strategy with the case of sufficient capital, the manufacturer's optimal production, carbon emission redcution and capital utilization policies are as follows:

(1) If $c_m < c_{m6}$, the manufacturer should offer the RIIA1 strategy that the initial capital of second period is abundant for production and R&D investment abatement.

(i) when $\tau 6 > \tau 5$, the strategy should satisfy the conditions that $c_m < c_{m7}$ and $\tau 6 < \tau < 1$, $\tau < \tau 7$.

(ii) when $\tau 6 < \tau 5$, the strategy should satisfy the conditions that $c_{m7} < c_m < c_{m6}$ and $\tau 5 < \tau < 1$, $\tau < \tau 7$.

The critical values of c_m are shown in Table 5, in which let $\tau 5 = \frac{n_2(2kq_{1n}\sigma - \omega_1)}{c_m^2 e_n^2 q_{1n}\sigma}$, $\tau 6 = \frac{2kq_{1n}\sigma - \omega_1}{2q_{1n}\sigma}$,

$$\tau 7 = \frac{c_m e_n (2kq_{1n}\sigma + c_n - Q) + 2n_2}{2c_m e_n q_{1n}\sigma},\quad c_{m6} = \sqrt{\frac{2n_2}{e_n^2}},\quad c_{m7} = \frac{Q - c_n - 2kq_{1n}\sigma}{e_n}$$

Proposition 3 characterizes the optimal strategy RIIA1 for the case of ample capital in the second

period by matching carbon price and the remanufacturing carbon intensity. Table 5 indicates that when the carbon price and the remanufacturing carbon intensity are constraint by certain values, the funds utilizing are independent of the initial capital in the second period.

Thus, the optimal total CERs can be given as: $E_{2a}^{(2)a} = \frac{c_m e_n{}^2 n_2 (\omega_1 - \omega_4)^2}{\omega_3{}^2}$

Table 5

The impact of c_m and τ on optimal strategies under R&D investment.

c_m		τ	The optimal strategies in period 2
$c_m < c_{m6} = \sqrt{\frac{2n_2}{e_n}}$	$c_m < c_{m7} = \frac{Q - c_n - 2kq_{1n}\sigma}{e_n}$	$\tau 6<\tau<1,\ \tau<\tau 7$	RIIA1
	$c_{m7} < c_m < c_{m6}$	$\tau 5<\tau<1,\ \tau<\tau 7$	
$c_m < c_{m6}$	$c_m < c_{m8} = \frac{Q - c_n - 2kq_{1n}\sigma - 2q_{2n}^{(2)b_2}}{e_n}$	$\tau 8<\tau<1,\ \tau<\tau 10$	RIIS1
	$c_{m8} < c_m < c_{m6}$	$\tau 9<\tau<1,\ \tau<\tau 10$	
The same as strategy RIIA1			RIIS2

(2) Insufficient capital and CQP

In this case, the initial capital of second period is insufficient and the remanufacturing firm pledges partial carbon quota for R&D abatement. Thus, the optimization model in this case can be formulated as:

$$max\,\Pi(q_{2n}^{(2)}, q_{2r}^{(2)}) = (p_{2n} - c_n)q_{2n} + (p_{2r} - c_r)q_{2r} - \varepsilon\sigma^2 - \frac{1}{2}n_2\tau_2{}^2 - c_m[(1-\tau_2)(e_n q_{2n} + e_r q_{2r}) - E_g] - rc_m\beta E_f \quad (6)$$

$$\text{s.t.} \begin{cases} c_n q_{2n} + c_r q_{2r} + \varepsilon\sigma^2 \le B_2 \\ \sigma q_{1n} = q_{2r} \\ c_n q_{2n} + c_r q_{2r} + \varepsilon\sigma^2 + \frac{1}{2}n_2\tau_2{}^2 = B_2 + c_m\beta E_f \\ 0 \le E_f \le E_g \\ q_{2n}, q_{2r} \ge 0 \end{cases}$$

The proofs for solving model (6) are omitted for brevity. Thus, we obtain the following Theorem:

Theorem 6. When the manufacturer conducts R&D investment abatement and the initial capital of the second period is insufficient, the optimal production decisions are shown in Table 4, which are significantly affected by c_m and τ. The critical of them define two decision strategies, namely RIIS1 strategy and RIIS2 strategy.

Theorem 6 explores two optimal strategies for the case of CQP in period 2. Associated with Theorem 6, we derive the Proposition 4 as follows:

Proposition 4. Under the RII strategy with the case of insufficient capital, the manufacturer's optimal production, CERs and capital utilization policies are as follows:

(1) If the manufacturer offers the RIIA1 strategy that the initial capital of second period is equal to production and collection and the quantity of carbon pledged is less than the carbon cap, then:

(i) when $\tau 8>\tau 9$, the strategy should satisfy the conditions that $c_m<c_{m8}$ and $\tau 8<\tau<1,\ \tau<\tau 10$.

(ii) when $\tau 8<\tau 9$, the strategy should satisfy the conditions that $c_{m8}<c_m<c_{m6}$ and $\tau 9<\tau<1$, $\tau<\tau 10$.

(2) If the manufacturer offers the RIIS2 strategy that not only the initial capital of second period is abundant but also the quantity of carbon pledged is less than the carbon cap, the critical parameters values are the same as strategy RIIA1 in the case of sufficient capital.

In Table 5, $c_{m8}=\dfrac{Q-c_n-2kq_{1n}\sigma-2q_{2n}^{(2)b_2}}{e_n}$, $\tau 8=\dfrac{\psi_3-\psi_4 q_{1n}+\psi_5}{c_n q_{1n}\sigma}$, $\tau 10=-\dfrac{\psi_3-\psi_4 q_{1n}+\psi_5}{c_n q_{1n}\sigma}-\dfrac{n_2}{c_m e_n q_{1n}\sigma}$,

$$\tau 9=-\frac{c_m{}^2 e_n{}^2\psi_3-2n_2 q_{1n}\sigma[c_n(k-1)+1]+\omega_3\psi_4 q_{1n}-\omega_3\psi_5-c_n n_2(c_m e_n-Q)}{c_n c_m{}^2 e_n{}^2 q_{1n}\sigma}.$$

Proposition 4 shows the optimal strategies for the situation of insufficient capital and carbon emission pledge in the second period. In Table 5, the critical parameters of c_m and τ constraint by certain values defining the case of insufficient capital as two strategies, RIIS1 and RIIS2. For strategy RIIS1, the initial capital in period 2 is equal to production and collection and we pledge partial carbon quota for R&D investment abatement. For strategy RIIS2, the initial capital of second period is abundant and the quantity of carbon pledged is less than the carbon cap, the optimal strategy is the same as the case in strategy RIIA1.

Thus, the optimal total CERs and the quantity of carbon pledged can be given as:

$$E_{2a}^{(2)b_2}=\frac{c_m e_n{}^2(\psi_3+c_n\tau q_{1n}\sigma-\psi_4 q_{1n}+\psi_5)^2}{c_n{}^2 n_2},\quad E_{2a}^{(2)b_3}=E_{2a}^{(2)a}$$

$$E_f^{(2)b_2}=\frac{c_m e_n{}^2(\psi_3+c_n\tau q_{1n}\sigma-\psi_4 q_{1n}+\psi_5)^2}{2c_n{}^2 n_2\beta}$$

$$E_f^{(2)b_3}=\frac{1}{c_m\beta}\{\frac{c_m{}^2 e_n{}^2 n_2(\omega_1-\omega_4)^2}{2\omega_3}+\frac{[n_2\omega_1-(2kn_2-c_m{}^2 e_n{}^2\tau)q_{1n}\sigma]c_n}{\omega_3}-\psi_3+q_{1n}\psi_4\}$$

5. Results and comparison

5.1. The comparison of two ways of CAI

Theorem 7 For the carbon purification abatement, when the carbon trading price increases($c_m\uparrow$), (1) the quantity of new and remanufacturing products in period 1 and period 2 will decrease ($q_{1n}^{(1)},q_{2n}^{(1)},q_{2r}^{(1)}\downarrow$), so are the CERs ($E_{1a}^{(1)},E_{2a}^{(1)}\downarrow$); (2) when the carbon trading price increases, the quantity of carbon pledged' monotonicity is not obvious ($E_f^{(1)}\downarrow\uparrow$), which need to be reviewed with graphs.

The detailed proof of Theorem 7 is provided in Appendix A.

Theorem 7 (1) suggests high carbon trading price may motivate the manufacturer to reduce carbon emissions, thus both the quantity of new and remanufactured products in period 1 and period 2 decrease, resulting in lower CERs. Theorem 7 (2) suggests the effect of carbon trading cost on the quantity of carbon pledged is uncertain. This finding offers a new sight on how to adjust quantity of new and remanufactured products and design for the appropriate quantity of carbon pledge by the manufacturer based on carbon price.

Example: we set e_n=4, σ=0.8, ξ=100, ϕ=0.3, B_1=1500, α=0.4, c_n=0.6, Δ=2, k=0.7, E_g=250, Q=200, τ=0.2, β=0.8, r=0.07 as the parameters to observe the results in Theorem 7. We vary the value of the carbon price c_m and other parameters remain unchanged. The curves of the decision variables in period 1 and period 2 are plotted in Figs. 2-3.

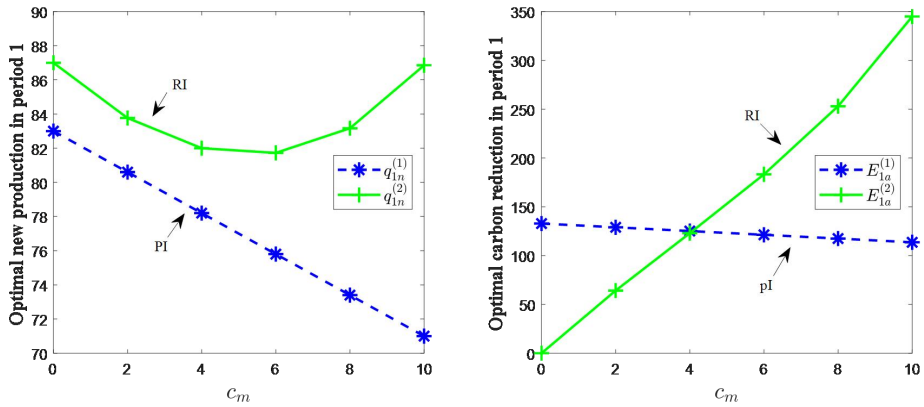

Fig. 2. Optimal production and carbon reduction in period 1 under two different carbon abatement investments

Figs. 2-3 illustrate the conclusion in Theorem 7. Figs. 2a-3b imply that when cost of carbon emission purification is greater than the carbon trading cost, there is no advantage to CER, thus the CERs is reduced. As the carbon trading cost c_m increases, the productions of new and remanufactured products reduce constrained by limited capital. Fig. 3c shows the quantity of carbon pledged stay the same first and decrease later. The reason behind it is when the carbon price less than a certain value, firm may pledge entire carbon cap for CER due to the limited carbon pledge quota. When the carbon price exceeds a certain value, firm may acquire more profits by selling carbon quotas, thus the quantity of carbon pledged decrease.

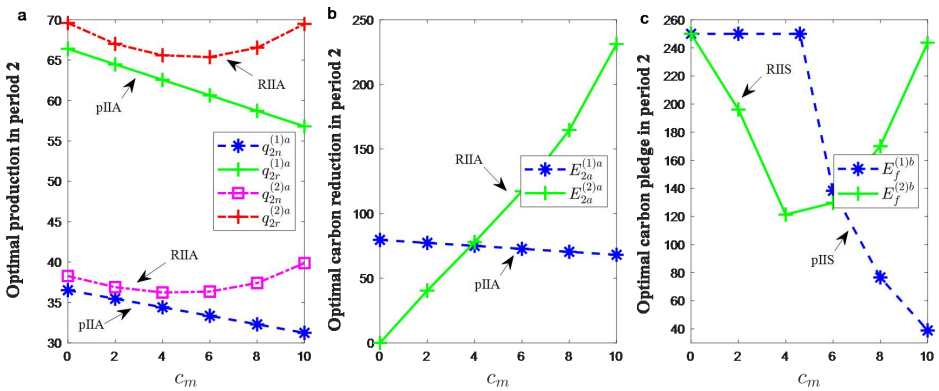

Fig. 3. Optimal production, carbon reduction and carbon pledge in period 2 under two different carbon abatement investments

Theorem 8 For the R&D investment abatement, when the carbon trading price increases ($c_m\uparrow$), (1) the monotonicity of new and remanufacturing production in period 1 and period 2 are not obvious, which need to be reviewed with graphs ($q_{1n}^{(2)}, q_{2n}^{(2)}, q_{2r}^{(2)} \downarrow\uparrow$), so is the quantity of carbon pledged ($E_f^{(2)} \uparrow\downarrow$); (2) the CERs in both period 1 and period 2 will decrease ($E_{1a}^{(2)}, E_{2a}^{(2)} \uparrow$).

A detailed proof of Theorem 8 is provided in Appendix A.

Suppose that n_1=3500, n_2=2000 (n_1>n_2) and we use the other general parameters set in Fig. 2 to observe the result of Theorem 8.

As illustrated in Figs. 2a-3b, for the R&D investment abatement, both the quantity of new and remanufactured products in period 1 and period 2 decrease first and increase later with the carbon price, while the CERs decrease. This is due to the R&D investment is proportional to the carbon price, higher

carbon price may motivate the manufacturer to conduct higher investment in R&D, resulting in higher CERs, thus the productions of new and remanufactured products decrease constrained by initial capital. Afterwards, as the carbon price goes up, the benefits of PCA are greater than the costs or benefits of variation in c_m, which may result in the increase in both PCAs. On the other hand, the quantity of carbon pledged decreases first and increases later in Fig. 3c, which monotonicity is not obvious, owing to the reason that the quantity of carbon pledged is related to changes of PCA. As c_m is lesser ($0<c_m\leq 4$), lower production leads to lower carbon emissions and lower carbon pledges; as c_m is larger ($c_m>4$), the trend of carbon pledge is similar with PCAs in Figs. 3a-3b. Hence, seen from the Theorem 7 and Theorem 8 above, the relationship between the carbon price and the quantity of carbon pledged implies that the method of carbon purification abatement or R&D investment abatement can effectively motivate manufacturer practice at different carbon price ranges.

Proposition 5 The remanufacturing firm under the situation of R&D investment produces more new products in period 1 (Fig. 2a) and produces more new and remanufacturing products in period 2 (Fig. 3a) compared with those under carbon purification.

Proposition 5 illustrates the R&D investment abatement is more effective in promoting recycling and remanufacturing compared to carbon purification abatement.

5.2. The impacts of ample capital for CQPF

（1）min B_2 for ample capital in II

Theorem 9 The minimum values of B_2 for ample capital in period 2 under carbon purification are a little higher than those under R&D investment with the self-owned capital B_1.

In Theorem 9, $\frac{\partial B_2^{(1)}}{\partial B_1}=\frac{\partial B_2^{(2)}}{\partial B_1}=1$, and it demonstrates that the R&D investment, comparing with the carbon purification, is more effective to achieve more initial capital in period 2 and decrease the possibility of pledge financing.

Example: we set c_p=6, e_n=3, σ=0.8, ξ=100, ϕ=0.35, c_m=3.5, α=0.5, c_n=5.5, Δ=2, k=0.7, E_g=250, Q=180 for the case of carbon purification, and suppose that n_1=3500, n_2=2000 and other parameters remain unchanged for the case of R&D investment to observe the results in Theorem 9. We vary the value of B_1 and other parameters unchanged. The curves of the minimum values of B_2 for ample capital in period 2 are plotted in Fig. 4.

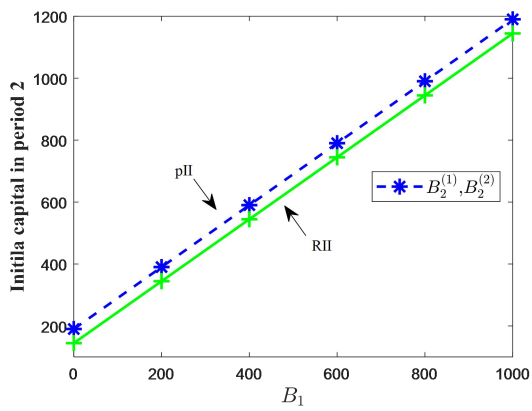

Fig. 4. The minimum values of B_2 for ample capital in period 2

Fig. 4 illustrates the conclusion in Theorem 9. It implies that the initial capital in period 2 are monotonic linear increasing with B_1 under both methods of CAI. It implies that in general the manufacturer will be motivated more easily to engage remanufacturing and CERs since the minimum values of B_2 for ample capital in period 2 is much higher under R&D investment rather than carbon purification.

（2）Critical τ for financing or not in period II

Theorem 10 The critical values of parameter τ for financing or not in period 2 will increase with B_1 under carbon purification abatement; while the critical values τ will decrease with B_1 under R&D investment abatement.

In Theorem 10, for the carbon purification abatement, $\frac{\partial \tau 1}{\partial B_1}=\frac{\partial \tau 1}{\partial B_1}=\frac{\partial \tau 3}{\partial B_1}=\frac{\partial \tau 4}{\partial B_1}=\frac{1}{c_p e_n q_{1n}\sigma}>0$; and for the R&D investment abatement, $\frac{\partial \tau 8}{\partial B_1}=\frac{\partial \tau 10}{\partial B_1}=-\frac{1}{c_n q_{1n}\sigma}<0$.

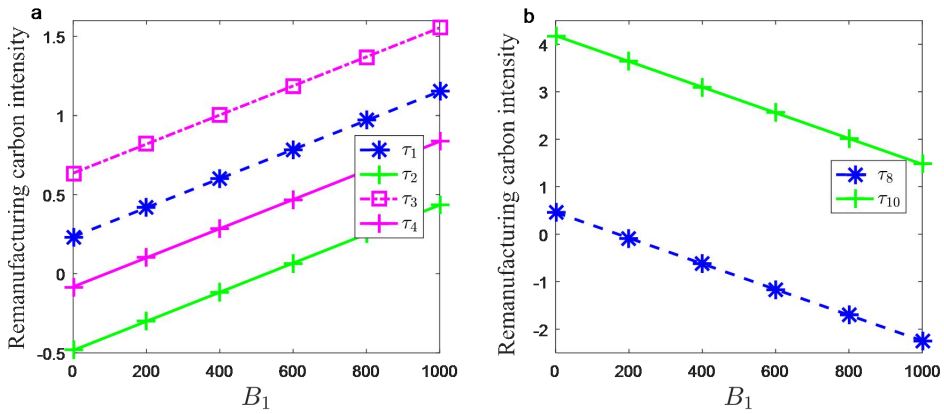

Fig. 5. The critical values of remanufacturing carbon intensity for financing or not

We use the parameters set in Fig. 4 to observe the results of Theorem 10. As illustrated in Fig. 5a, the critical values of parameter τ ($\tau 1$-$\tau 4$) are monotonic increasing with the self-owned capital in period 1, which reflects that the optional range of remanufacturing carbon intensity will be wider when B_1 increase under the situation of carbon purification abatement. While on the contrary, the optional range $\tau 8$ or $\tau 10$ for remanufacturing carbon intensity narrows when B_1 increase under R&D investment abatement in Fig. 5b. Hence, the relationship between the remanufacturing carbon intensity and the initial capital in period 1 implies that the difference of the optional range of remanufacturing carbon intensity may urge manufacturer set reasonable parameter values according to different pledge financing choices.

6. Numerical examples and sensitivity analysis

In this section, we use numerical example to comparative discuss the impact of carbon purification and R&D investment abatements on firm's optimal PCA strategies in period 2, as well as the impact of initial capital in period 1 on firm's total carbon emissions and profits under two different CAI cases. Assume that the carbon caps allocated by the government and market demands in two periods are the same, that is $Eg_1=Eg_2=Eg$, $Q_1=Q_2$.

6.1. Sensitivity to the carbon trading price c_m

The fact that the carbon trading price per unit in the market changes dynamically, thus, in this subsection, the sensitivity to the carbon price is analyzed to explore the relationship among the two CER cases. We vary the value of c_m (from 0 to 10) in period 2 under two fixed values of α for the case of carbon purification and two sets of fixed values of n_1 and n_2 for the case of R&D investment, while keeping all other base parameters set in Theorem 7 unchanged. Thus, Figs. 6-7 illustrate the impact of c_m on the optimal decisions in the second period under two different CAI cases.

Fig. 6 depicts the variation of the differences between carbon purification and R&D investment in production, CER and profits when the initial capital is sufficient in the second period. It shows that the increments with the first set of parameter values ($\alpha_1=0.4$, n_{11}=3500, n_{21}=2000) are always higher than those with the second set of parameters ($\alpha_2=0.8$, n_{12}=4000, n_{22}=2500) with the increasing c_m. Fig. 6a displays the differences in the optimal production of new and remanufactured products in period 2 with different parameter values always positive apart from the optimal new production quantity with values α_2, n_{12} and n_{22}. It means the quantities of new and remanufactured products under R&D investment usually overweight those under carbon purification. The variation trends of the differences decrease first and increase later with c_m, while the difference curves present a comparatively rapid growth trend with the first set of parameters, which means the carbon purification gains a higher advantage over the R&D investment as the parameters become smaller.

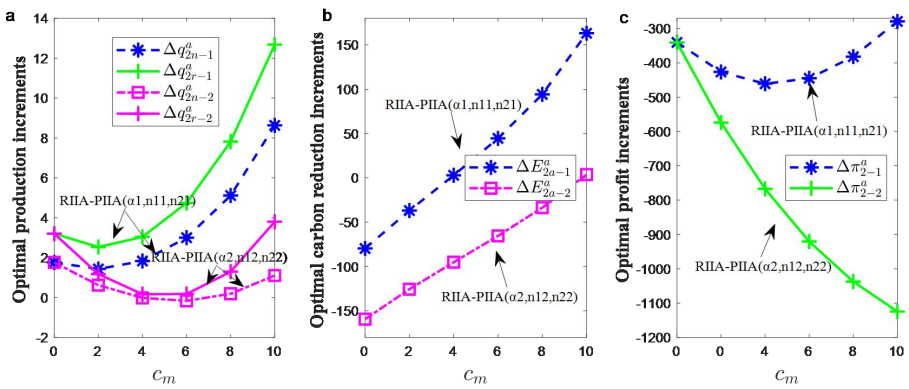

Fig. 6. The optimal production, carbon and profit increments of sufficient capital in the second period with different parameters of α, n_1 and n_2

The curves depicted in Fig. 6b shows that the CERs under R&D investment is lower compared with the case under carbon purification when $c_m \leq 4$ in this example and may be higher when $c_m>4$ (α_1, n_{11}, n_{21}). On the other hand, the CERs under carbon purification always get larger in this example (α_2, n_{12}, n_{22}). Fig. 6c displays the optimal profit under R&D investment always overweighs the case under carbon purification with different parameter values in this example, and the variation trend decreases first and increase later with the parameters α_1, n_{11} and n_{21} with c_m, while the curve presents a logarithmic downward trend with parameters α_2, n_{12} and n_{22}.

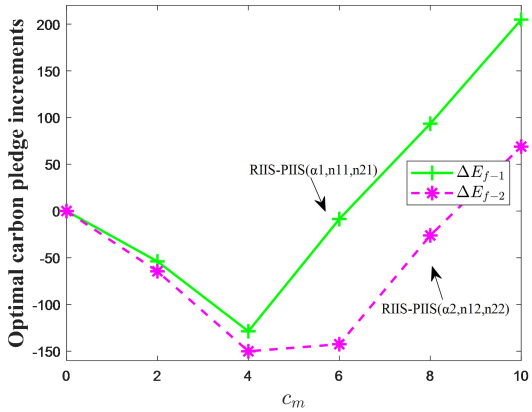

Fig. 7. The optimal carbon pledge increments of insufficient capital in the second period with different parameters of α, n_1 and n_2

The difference curves between carbon purification and R&D investment in production, production, CER and profits with insufficient initial capital are similar with those when the initial capital is sufficient in the second period, here no longer say. The curves depicted in Fig. 7 indicate that the manufacturer pledge less carbon quota under R&D investment compared with that under carbon purification as $c_m \leq 4$ and pledge more carbon quota as $c_m > 4$ in this example (α_1, n_{11}, n_{21}). While the manufacturer pledge more carbon quota under R&D investment for a larger c_m ($c_m > 8.5$) with the second parameter values. Specially, the trends of the difference of carbon pledge quantity get approximate straight-line go down first and then approximate linear go up with different parameter values with c_m, which all present a lowest point in the curves.

6.2. Sensitivity to the self-owned capital B_1 of first period

In this subsection, the sensitivity to the self-owned capital in the first period is analyzed. Based on the parameters set in Theorem 8, we assumed that the carbon price c_m=3.5 and we vary the value of B_1 (from 0 to 1000) in the following analysis. Figs. 8-10 illustrate the impact of self-owned capital in period 1 on the CERs, carbon pledges and profits in the second period of sufficient initial capital or insufficient initial capital under two cases of R&D investment and carbon purification.

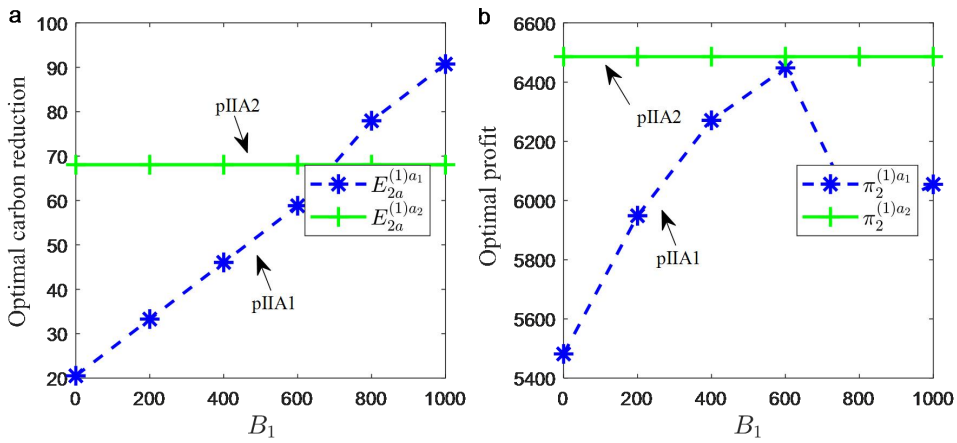

Fig. 8. The optimal carbon reduction and profit of sufficient capital in the second period under carbon purification

Figs. 8-9 depict the two possible solutions for sufficient capital (a_1, a_2) or insufficient capital (b_1, b_3) in the second period respectively under carbon purification. In the situation of sufficient capital (Fig. 8), the CER and profit of solution a_1 almost unchanged, while the CER is rising and the profit presents a trend of rising first and then falling and rising again of solution a_2. The behavior can be explained by the fact that the firm use most of the initial capital for production for a smaller B_1, resulting the CER and profit growth in period 2. As B_1 continues to grow, the production quantities of new and remanufactured products gradually reached the maximum under the present conditions, and then the CER increase makes the total profit dip a few. Later, continued increase in CER will obtain more carbon trading revenue for the firm, which brings more overall profit in period 2.

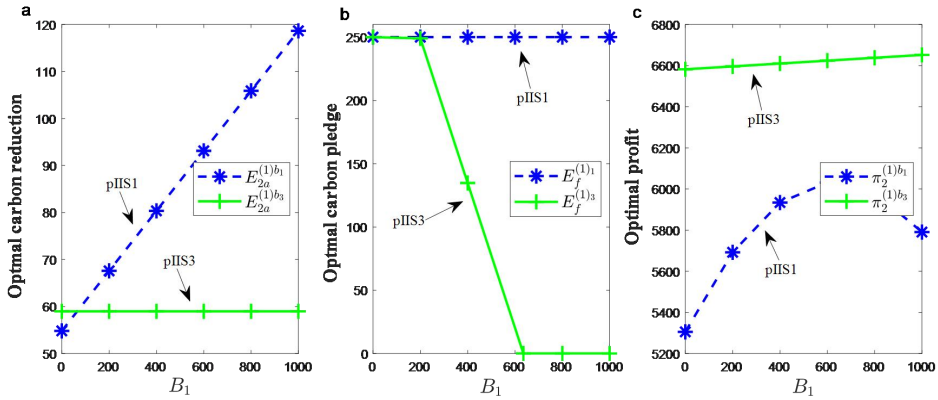

Fig. 9. The optimal carbon reduction, carbon pledge and profit of insufficient capital in the second period under carbon purification

In the case of insufficient capital, the quantity of carbon pledged of situation b_3 keeping at the highest value first, and then fall to zero, and finally remain unchanged with B_1. It can be implied that the firm pledge a large amount of carbon quota for CERs as B_1 is small, resulting in a rapid growth in production in period 2. As B_1 continues to grow, the production quantity gradually reached the maximum, which makes the carbon pledges dip rapidly. Finally, B_1 just meet the double need of PCA and pull the carbon emission pledges to zero. For another, the trend of CER and profit of situation b_1 can be explained as the same reason as the case of sufficient capital, here no longer etc., and the quantity of carbon pledged of situation b_1 may be irrelevant with B_1.

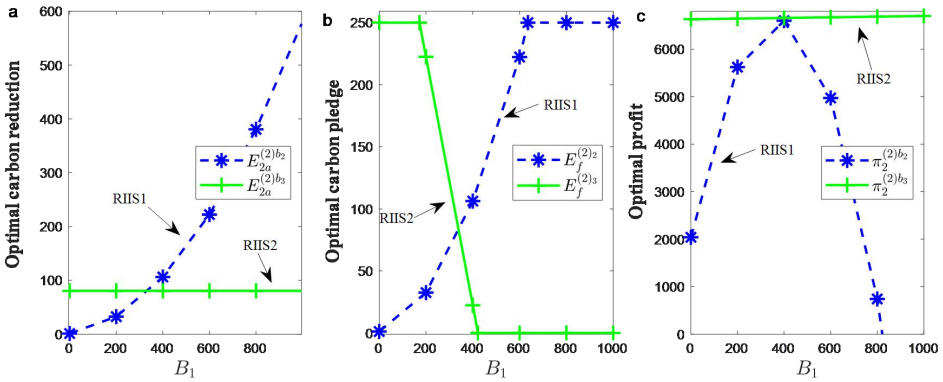

Fig. 10. The optimal carbon reduction, carbon pledge and profit of insufficient capital in the second period under R&D investment

Fig. 10 only depicts the two possible solutions for insufficient capital (b_2, b_3) in the second period under R&D investment, while the decision variables of sufficient capital are independent of the change

in B_1. In Fig. 10, the variation trend of CER of situation b_2 is similar with that of situation b1 in Fig. 9a, and the profit of situation b_2 presents a trend of rapid growth first and decline later with B_1. The reason behind it can be explained as the same as that of situation b_1 under carbon purification (Fig. 9c). Nevertheless, the production, CER effect enhances due to the cost coefficient of CER getting smaller in the second period. Thus, the CER increases rapidly, which leads to a quickly decline of the total profit after it reaches the maximum. The curve of carbon pledges of situation b_2 in Fig. 10b illustrates that with a smaller cost coefficient of CER in the second period, the firm pledge a large amount of carbon quota for CER when B_1 is small. As B_1 continues to grow, the production quantity gradually reached the maximum, which makes the carbon pledges reaching the peak value and remain unchanged.

7. Discussion and conclusions

This paper explores the integrated optimization of PCA decisions in a two-period manufacturing/remanufacturing system, in which the firm produces new products in the first period and produces new and remanufactured products simultaneously in the second period. To support CERs, the firm pledge partial carbon cap allocated by the government for financing, redeeming the carbon quota at the end of the period. Accordingly, different optimization models and analytical expressions based on profit are established and numerical experiments are set up to contrastively evaluate the effect of carbon pledge financing on the optimal decisions under two CAI methods.

From a managerial point of review, carbon pledge financing more effectively enhance CERs and increase profits of firms under two CAI when the self-owned capital is shortage, whereas it contributes to the increase in carbon emissions simultaneously. In addition, insufficient initial capital in the first stage would indirectly affect the production decisions of the second stage and the unit carbon trading price can affect the financing decision of remanufacturing firm at this stage. We also find that R&D investment has a cumulative effect, namely, the CER effect of the second period is more obvious than that of the first period, and the CERs and expected profits may be higher than the case of carbon emission purification.

To the best of our knowledge, this paper can be expended in several directions. Firstly, apart from carbon emission purification and R&D investment, a few other ways to reduce emissions and adopting multiple methods of CER simultaneously are worth further exploring. In addition, CQPF can flexibly set the repayment period and loan amount according to CERs. Other results on the operational and financial implications presented in this paper may also serve as hypotheses for further empirical researches.

CRediT authorship contribution statement

Qianru Li: Conceptualization, Methodology, Writing-original draft, Funding acquisition. **Xinxin Guo**: Software, Formal analysis, Funding acquisition, Writing-original draft. **Lu Chen**: Conceptualization, Writing-review & editing. **Mengxue Yang**: Writing-review & editing, Conceptualization, Methodology.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper

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Data availability

Data will be made available on request.

Appendix A

A.1. Proof of Theorem 1.

To show that the objective function (1) is concave, we need to prove that its second derivative with respect to the new production is negative (see A.2). Furthermore, the initial cash constraint is also differentiable, and it is concave in q_{1n}. The first and the second derivative of the objective function (1) are given below:

$$\frac{\partial \prod(q_{1n}^{(1)})}{\partial q_{1n}} = -2q_{1n} + Q - c_n - c_p e_n - c_m(1-\alpha)e_n \tag{A.1}$$

$$\frac{\partial^2 \Pi(q_{1n}^{(1)})}{\partial q_{1n}^2} = -2 \tag{A.2}$$

It is obvious that the second derivative is negative the objective function (1) is concave. Thus, we derive the optimal solution of new production $q_{1n}^{(1)}$ by taking advantage of the Karush-Kuhn-Tucker (KKT) conditions below.

$$L(q_{1n}) = (Q - q_{1n} - c_n)q_{1n} - c_p e_n q_{1n} - c_m((1-\alpha)e_n q_{1n} - E_g) - \lambda_1(-c_p e_n q_{1n} - c_n q_{1n} + B_1) \tag{A.3}$$

$$\begin{cases} -2q_{1n} + Q - c_n - c_p e_n - c_m(1-\alpha)e_n - \lambda_1(-c_p e_n - c_n) = 0 \\ \lambda_1(-c_p e_n q_{1n} - c_n q_{1n} + B_1) = 0 \\ -c_p e_n q_{1n} - c_n q_{1n} + B_1 > 0 \\ \lambda_1 \geq 0 \end{cases} \tag{A.4}$$

We derive one possible case, thus, there is only one feasible solution $q_{1n}^{(1)}$, that is:

$$q_{1n}^{(1)} = \frac{1}{2}[Q - c_p e_n - c_n - c_m e_n(1-\alpha)]$$

$q_{1n}^{(1)} > 0$, we can get that $c_m < c_{m1} = \dfrac{\psi_2}{e_n(1-\alpha)}$

Among, $\psi_1 = c_m e_n(1-\alpha)$, $\psi_2 = Q - c_p e_n - c_n$.

A.2. Proof of Theorem 2 and proposition 1.

It can be verified that objective function (2) and the initial cash constraint are strictly concave in q_{2n} and q_{2r}. The optimality conditions of Model (2) can be characterized by Karush-Kuhn-Tucker (KKT)

conditions. We use λ_1, λ_2 as the multiplier corresponding to the slack variable, the Lagrangian function of Model (1) can be written as following:

$$\begin{aligned} L(q_{2n}) = & (-kq_{1n}\sigma + Q - c_n - q_{2n})\, q_{2n} + (k(-q_{1n}\sigma + Q - q_{2n}) - c_n + \Delta)\sigma q_{1n} - \xi\sigma^2 - c_p(e_n q_{1n}\sigma\tau + e_n q_{2n}) \\ & - c_m((1-\alpha)(e_n q_{1n}\sigma\tau + e_n q_{2n}) - E_g) + \lambda_1(-\xi\sigma^2 - c_p(e_n q_{1n}\sigma\tau + e_n q_{2n}) - c_n q_{2n} - (c_n - \Delta)\sigma q_{1n} + B_2) \end{aligned} \tag{A.5}$$

$$\begin{cases} -2k\sigma q_{1n} + Q - c_n - 2q_{2n} - c_p e_n - c_m(1-\alpha)e_n + \lambda_1(-c_p e_n - c_n) = 0 \\ \lambda_1(-\xi\sigma^2 - c_p(e_n q_{1n}\sigma\tau + e_n q_{2n}) - c_n q_{2n} - (c_n - \Delta)\sigma q_{1n} + B_2) = 0 \\ -\xi\sigma^2 - c_p(e_n q_{1n}\sigma\tau + e_n q_{2n}) - c_n q_{2n} - (c_n - \Delta)\sigma q_{1n} + B_2 \geq 0 \\ \lambda_1 \geq 0 \end{cases} \tag{A.6}$$

Depending on whether the multipliers λ_1, λ_2 are zero or positive, we derive four possible cases. However, there are only two cases that may result in feasible solutions.

Case 1 when λ_1>0, λ_2=0, the initial cash of the manufacturer is sufficient for recycling, production and carbon emission purification (strategy PIIA1). Combing Eq. (2) and Eq. (A.5)—(A.6), we have

$$q_{2n}^{(1)} = \frac{\psi_3 - c_p e_n \tau q_{1n}\sigma - \psi_4 q_{1n} + \psi_5}{\psi_4} \tag{A.7}$$

$$q_{2r}^{(1)} = \sigma q_{1n}^{(1)} = \frac{1}{2}\sigma(\psi_2 - \psi_1) \tag{A.8}$$

$$\lambda_1^{(1)} = -\frac{\psi_1\psi_4 + 2kq_{1n}\sigma\psi_4 - 2c_p e_n q_{1n}\sigma\tau + 2\psi_3 - \psi_2\psi_4 - 2\sigma^2\xi + 2B_2}{\psi_4^{\ 2}} \tag{A.9}$$

From $q_{2n}^{(1)} > 0$ *and* $q_{2r}^{(1)} > 0$, we can get $c_m < c_{m1}$ and $\tau < \frac{\psi_3 - \psi_4 q_{1n} + \psi_5}{c_p e_n q_{1n}\sigma} = \tau_1$. Also, form $\lambda_1^{(1)} \geq 0$, we can obtain that $\tau > \frac{\psi_4[2kq_{1n}\sigma + (1-\alpha)c_m e_n - \psi_2 - 2q_{1n}]) + 2\psi_3 + 2\psi_5}{2c_p e_n q_{1n}\sigma} = \tau_2$

$$\tau_1 - \tau_2 = \frac{2\psi_4(1-k\sigma)q_{1n}}{2c_p e_n q_{1n}\sigma} > 0 \quad , \quad \tau_2 < \tau < \tau_1 \tag{A.10}$$

Among, $\psi_3 = (\Delta - c_n)q_{1n}\sigma$, $\psi_4 = c_p e_n + c_n$, $\psi_5 = -\varepsilon\sigma^2 + \phi\pi_1 + B_1$

Based on the value range of all parameters, when $c_m < c_{m1}$ and $\tau_2 < \tau < \tau_1$ (Table 2) are satisfied, the values of $q_{2n}^{(1)a_1}$, $q_{2r}^{(1)a_1}$ in this case are the optimal results, the conclusion of line 1 in Table 1 is proved.

Case 2 when λ_1=0, λ_2=0, the initial cash of the manufacturer is just right equal to the cost of recycling, production and carbon emission purification (strategy PIIA2). Combing Eq. (2) and Eq. (A.5)—(A.6), we have:

$$q_{2n}^{(1)} = \frac{1}{2}(\psi_2 - \psi_1) - k\sigma q_{1n} \tag{A.11}$$

$$q_{2r}^{(1)} = \sigma q_{1n}^{(1)} \tag{A.12}$$

$q_{2n}^{(1)} > 0$ *and* $q_{2r}^{(1)} > 0$ should be satisfied, thus we can obtain that $c_m < c_{m1}$

In this case, Eq. (A.11) and Eq. (A.12) must be positive, thus, the necessary condition for this case is: $c_m < c_{m1}$, we can derive the optimal results $q_{2n}^{(1)a_2}$ and $q_{2r}^{(1)a_2}$, the conclusion of line 2 in Table 1 is proved.

A.3. Proof of Theorem 3 and proposition 2.

It can be verified that objective function (3) and the initial cash constraint are strictly concave in q_{2n} and q_{2r}. And the Lagrangian and the Karush-Kuhn-Tucker optimality conditions of Model (2) are similar to those of Theorem 2, we omit it here.

The multipliers λ_1, λ_2 may be equal or greater than zero. Considering all these alternatives, we derive four possible cases, however, there are three cases that may result in feasible solutions.

Case 1 when λ_1>0, λ_2=0, the initial cash of the manufacturer is sufficient for recycling and production and the quantity of carbon quota pledged are less than the carbon cap (strategy PIIS1). We have:

$$q_{2n}^{(1)} = \frac{\psi_3 - c_p e_n \tau q_{1n}\sigma - \psi_4 q_{1n} + \psi_5 + E_g \beta c_m}{\psi_4} \tag{A.13}$$

$$q_{2r}^{(1)} = \sigma q_{1n}^{(1)} \tag{A.14}$$

$$\lambda_1^{(1)} = -\frac{c_n\psi_1 + 2c_n k q_{1n}\sigma + 2\psi_3 - c_n\psi_2 - 2q_{1n}\psi_4 - 2\sigma^2\xi + 2\phi\pi_1 + 2B_1}{c_n^{\ 2}} \tag{A.15}$$

From $q_{2n}^{(1)} > 0 \ and \ q_{2r}^{(1)} > 0$, we can get that $c_m < c_{m1}$ and $\tau < \frac{\psi_3 - \psi_4 q_{1n} + \psi_5 + E_g \beta c_m}{c_p e_n q_{1n}\sigma} = \tau_3$. Also, from $\lambda_1^{(1)} \geq 0$, we can obtain that $\tau > \frac{\psi_4[2kq_{1n}\sigma + (1-\alpha)c_m e_n - \psi_2 - 2q_{1n}]) + 2\psi_3 + 2\psi_5 + 2E_g\beta c_m}{2c_p e_n q_{1n}\sigma} = \tau_4$

$$\tau_3 - \tau_4 = \frac{2\psi_4(1-k\sigma)q_{1n}}{2c_p e_n q_{1n}\sigma} > 0 \quad , \tau_4 < \tau < \tau_3 \tag{A.16}$$

Based on the value range of all parameters, when $c_m < c_{m1}$ and $\tau_4 < \tau < \tau_3$ (Table 2) are satisfied, the values of $q_{2n}^{(1)b_1}$, $q_{2r}^{(1)b_1}$ in this case are the optimal results, the conclusion of line 3 in Table 1 is proved.

Case 2 when λ_1=0, λ_2=0, the initial cash of the manufacturer is just right equal to the cost of recycling and production and the quantity of carbon quota pledged are just equal to the carbon cap (strategy PIIS2). Then：

$$q_{2n}^{(1)} = \frac{\psi_3 - \psi_4 q_{1n} + \psi_5}{c_n} \tag{A.17}$$

$$q_{2r}^{(1)} = \sigma q_{1n}^{(1)} \tag{A.18}$$

$q_{2n}^{(1)} > 0 \ and \ q_{2r}^{(1)} > 0$ should be satisfied, thus from $q_{2n}^{(1)b_2} - q_{1n}^{(1)} = -k\sigma q_{1n}^{(1)}$, $q_{2n}^{(1)b_2} = (1-k\sigma)q_{1n}^{(1)} > 0$, we can get the condition that $c_m < c_{m1}$.

In this case, Eq. (A.17) and Eq. (A.18) must be positive, thus, the necessary condition for this case is $c_m < c_{m1}$, we can derive the optimal results $q_{2n}^{(1)b_2}$ and $q_{2r}^{(1)b_2}$, the conclusion of line 4 in Table 1 is proved.

Case 3 when λ_1=0, λ_2>0, the initial cash of the manufacturer is just right equal to the cost of recycling and production and the quantity of carbon quota pledged are just equal to the carbon cap (strategy PIIS3)：

$$q_{2n}^{(1)} = \frac{1}{2}[\psi_2 - \psi_1 - k\sigma q_{1n}] \tag{A.19}$$

$$\lambda_2^{(1)} = -\frac{c_m\beta(\psi_1\psi_4 + 2kq_{1n}\sigma\psi_4 - 2c_p e_n q_{1n}\sigma\tau - \psi_2\psi_4 + 2\psi_3 + 2E_g\beta c_m - 2\sigma^2\xi + 2B_2)}{\psi_4^{\ 2}} \quad \text{(A.20)}$$

From $q_{2n}^{(1)} > 0$ *and* $q_{2r}^{(1)} > 0$, we can get that $c_m < c_{m1}$. Also, from $\lambda_2^{(1)} \geq 0$, we should satisfy that

$$c_m < \frac{1-\psi_4 - 2kq_{1n}\sigma - 2q_{2n}^{(2)b_2}}{e_n(1-\alpha)} = c_{m2}$$

$c_{m2} - c_{m1} = -\dfrac{2c_n(kq_{1n}\sigma + q_{3n2})}{c_m e_n(1-\alpha)} < 0$, thus $c_m < c_{m2}$

Based on the value range of all parameters, when $c_m < c_{m2}$ (Table 2) are satisfied, the values of $q_{2n}^{(1)b_3}$, $q_{2r}^{(1)b_3}$ in this case are the optimal results, the conclusion of line 5 in Table 1 is proved.

For carbon emission purification, substituting the respective optimal production quantities of new and remanufactured products into the objective function, we obtain the optimal profit of the manufacturer for each case.

A.4. Proof of Theorem 4.

It can be verified that objective function (4) and the initial cash constraint are strictly concave in q_{1n} *and* τ_1. We can derive the optimal solutions through conducting the Karush-Kuhn-Tucker (KKT) conditions to ensure the global optimality together with the feasible conditions:

$$L(q_{1n},\tau_1) = (Q - q_{1n} - c_n)q_{1n} - \frac{1}{2}n_1\tau_1^{\ 2} - c_m((1-\tau_1)e_n q_{1n} - E_g) - \lambda_1(B_1 - c_n q_{1n} - \frac{1}{2}n_1\tau_1^{\ 2}) \quad \text{(A.21)}$$

$$\begin{cases} -2q_{1n} + Q - c_n - c_m(1-\tau_1)e_n + \lambda_1 c_n = 0 \\ c_m e_n q_{1n} + \lambda_1 n_1\tau_1 - n_1\tau_1 = 0 \\ \lambda_1(B_1 - c_n q_{1n} - \frac{1}{2}n_1\tau_1^2) = 0 \\ B_1 - c_n q_{1n} - \frac{1}{2}n_1\tau_1^{\ 2} > 0 \\ \lambda_1 \geq 0 \end{cases} \quad \text{(A.22)}$$

We derive only one feasible solution of $q_{1n}^{(2)}$ and $\tau_1^{(2)}$, that is:

$q_{1n}^{(2)} = \dfrac{n_1\omega_1}{\omega_2}$ and $\tau_1^{(2)} = \dfrac{c_m e_n \omega_1}{\omega_2}$

From $q_{1n}^{(2)} > 0$, $\tau_1^{(2)} > 0$, we can obtain that $c_m < \dfrac{1-c_n}{e_n} = c_{m3}$ and $c_m < \sqrt{\dfrac{2n_1}{e_n^{\ 2}}} = c_{m4}$

In order to satisfy $\tau_1^{(2)} < 1$, we can get that $c_m > \dfrac{2n_1}{e_n(Q-c_n)} = c_{m5}$

Among, $\omega_1 = Q - c_n - c_m e_n$, $\omega_2 = 2n_1 - c_m^{\ 2} e_n^{\ 2}$

Based on the value range of all parameters, when $c_m < c_{m3}$, $c_m < c_{m4}$ and $c_m > c_{m5}$ are satisfied, the values of $q_{1n}^{(2)}$ and $\tau_1^{(2)}$ in this case are the optimal results.

A.5. Proof of Theorem 5 and proposition 3.

It can be verified that objective function (5) and the initial cash constraint are strictly concave in q_{2n}, q_{2r} and τ. Thus, the Lagrangian and the Karush-Kuhn-Tucker optimality conditions of Model (1) are as

follows:

$$L(q_{2n},\tau_2)=(-kq_{1n}\sigma+Q-c_n-q_{2n})q_{2n}+(k(-q_{1n}\sigma+Q-q_{2n})-c_n+\Delta)\sigma q_{1n}-\xi\sigma^2-\frac{1}{2}n_2\tau_2{}^2$$
$$-c_m((1-\tau_2)(e_nq_{1n}\sigma\tau+e_nq_{2n})-E_g)+\lambda_1(B_2-c_nq_{2n}-(c_n-\Delta)\sigma q_{1n}-\xi\sigma^2-\frac{1}{2}n_2\tau_2{}^2) \quad (A.23)$$

$$\begin{cases}-2k\sigma q_{1n}+Q-c_n-2q_{2n}-c_m(1-\tau_2)e_n-\lambda_1c_n=0\\ -n_2\tau_2-c_m(-e_nq_{1n}\sigma\tau-e_nq_{2n})-\lambda_1n_2\tau_2=0\\ B_2-c_nq_{2n}-(c_n-\Delta)\sigma q_{1n}-\xi\sigma^2-\frac{1}{2}n_2\tau_2{}^2=0\\ \lambda_1(B_2-c_nq_{2n}-(c_n-\Delta)\sigma q_{1n}-\xi\sigma^2-\frac{1}{2}n_2\tau_2{}^2)\geq 0\\ \lambda_1\geq 0\end{cases} \quad (A.24)$$

Considering all these alternatives, we derive only one case that may result in the feasible solution, that is: λ_1=0.

When λ_1=0, the initial cash of the manufacturer is sufficient for recycling, production and R&D investment abatement (strategy RIIA1). Combing Eq. (5) and Eq. (A.23)—(A.24), we have:

$$q_{2n}^{(2)}=\frac{n_2\omega_1-(2kn_2-c_m{}^2e_n{}^2\tau)q_{1n}\sigma}{\omega_3} \quad (A.25)$$

$$q_{2r}^{(2)}=\sigma q_{1n}^{(2)} \quad (A.26)$$

$$\tau_2^{(2)}=\frac{c_me_n[\omega_1+\omega_4]}{\omega_3} \quad (A.27)$$

From $q_{2n}^{(2)}>0, q_{2r}^{(2)}>0$ and $\tau_2^{(2)}>0$, we can obtain that $\tau>\frac{n_2(2kq_{1n}\sigma-\omega_1)}{c_m{}^2e_n{}^2q_{1n}\sigma}=\tau_5$ and $\tau>\frac{2kq_{1n}\sigma-\omega_1}{2q_{1n}\sigma}=\tau_6$. Also, from $\tau_2^{(2)}<1$, we can get that $\tau<\frac{c_me_n(2kq_{1n}\sigma+c_n-Q)+2n_2}{2c_me_nq_{1n}\sigma}=\tau_7$

In order to satisfy $\tau_2^{(2)a}<1$, $\tau<\tau_7=\frac{c_me_n(2kq_{1n}\sigma+c_n-Q)+2n_2}{2c_me_nq_{1n}\sigma}$

① If $\tau_5-\tau_6=\frac{(2kq_{1n}\sigma-\omega_1)\omega_3}{2c_m{}^2e_n{}^2q_{1n}\sigma}>0$, thus $c_{m7}<c_m<c_{m6}\Rightarrow\tau_5<\tau<1$ and $\tau<\tau_7$

② If $\tau_5-\tau_6<0$, thus $c_m<c_{m7}\Rightarrow\tau_6<\tau<1$ and $\tau<\tau_7$

Among, $\omega_3=2n_2-c_m{}^2e_n{}^2$, $\omega_4=2q_{1n}\sigma(\tau-k)$, $c_{m6}=\sqrt{\frac{2n_2}{e_n{}^2}}$ $c_{m7}=\frac{Q-c_n-2kq_{1n}\sigma}{e_n}$

Based on the value range of all parameters, if $\tau_5-\tau_6>0$, $c_{m7}<c_m<c_{m6}$, $\tau_5<\tau<1$ and $\tau<\tau_7$ are satisfied (Table 4), we can derive the optimal solutions of $q_{2n}^{(2)a_1}, q_{2r}^{(2)a_1}$ and $\tau_2^{(2)a_1}$. Otherwise, we can derive the optimal results by satisfying the conditions of $c_m<c_{m7}$, $\tau_6<\tau<1$ and $\tau<\tau_7$, the conclusion of line 1 in Table 3 is proved.

A.6. Proof of Theorem 6 and proposition 4.

It can be verified that objective function (6) and the initial cash constraint are strictly concave in q_{2n},

q_{2r} and τ. Thus, the Lagrangian and the Karush-Kuhn-Tucker optimality conditions of Model (1) are similar to those of Theorem 5, we omit it here.

Depending on whether the multipliers λ_1, λ_2 are zero or positive, we derive two cases that may result in the feasible solutions:

Case 1 when λ_1>0, λ_2=0, the initial cash of the manufacturer is sufficient for recycling and production, and the quantity of carbon quota pledged are just equal to carbon cap (strategy RIIS1). We have:

$$q_{2n}^{(2)} = \frac{\psi_3 - \psi_4 q_{1n} + \psi_5}{c_n} \tag{A.28}$$

$$\tau_2^{(2)} = \frac{c_m e_n(\psi_3 + c_n \tau q_{1n}\sigma) - \psi_4 q_{1n} + \psi_5}{c_n n_2} \tag{A.29}$$

$$\begin{aligned}\lambda_1^{(2)} = -1 + &\frac{c_m{}^2 e_n{}^2 q_{1n}(\sigma\tau - \sigma - 1) - c_m e_n n_2 + 2n_2 q_{1n}\sigma(1-k) + n_2(Q + 2q_{1n})}{n_2 c_n} \\ &+ \frac{(c_m{}^2 e_n{}^2 - 2n_2)(\Delta q_{1n}\sigma - \sigma^2\xi - c_p e_n q_{1n} + \phi\pi_1 + B_1)}{n_2 c_n{}^2}\end{aligned} \tag{A.30}$$

From $q_{2n}^{(2)} > 0, q_{2r}^{(2)} > 0$ and $\tau_2^{(2)} > 0$, we can get that $\tau > \frac{\psi_3 - \psi_4 q_{1n} + \psi_5}{c_n q_{1n}\sigma} = \tau_8$, also, from $\lambda_1^{(2)} > 0$, we have $\tau > -\frac{c_m{}^2 e_n{}^2 \psi_3 - 2n_2 q_{1n}\sigma[c_n(k-1)+1] + \omega_3\psi_4 q_{1n} - \omega_3\psi_5 - c_n n_2(c_m e_n - Q)}{c_n c_m{}^2 e_n{}^2 q_{1n}\sigma} = \tau_9$. In order to satisfy $\tau_2^{(2)b} < 1$, $\tau < -\frac{\psi_3 - \psi_4 q_{1n} + \psi_5}{c_n q_{1n}\sigma} - \frac{n_2}{c_m e_n q_{1n}\sigma} = \tau_{10}$

① If $\tau_8 - \tau_9 > 0$, thus $c_m < c_{m8} = \frac{Q - c_n - 2kq_{1n}\sigma - 2q_{2n}^{(2)b_2}}{e_n}$, $\tau_8 < \tau < 1$ and $\tau < \tau_{10}$

② If $\tau_8 - \tau_9 < 0$, thus $c_{m8} < c_m < c_{m6}$, $\tau_9 < \tau < 1$ and $\tau < \tau_{10}$

Based on the value range of all parameters, if $\tau_8 - \tau_9 > 0$, $c_m < c_{m8}$, $\tau_8 < \tau < 1$ and $\tau < \tau_{10}$ are satisfied (Table 4), we can derive the optimal solutions of $q_{2n}^{(2)b_1}, q_{2r}^{(2)b_1}$ and $\tau_2^{(2)b_1}$. Otherwise, we can derive the optimal results by satisfying the conditions of $c_{m8} < c_m < c_{m6}$, $\tau_9 < \tau < 1$ and $\tau < \tau_{10}$, the conclusion of line 2 in Table 3 is proved.

Case 2 when λ_1=0, λ_2=0, the initial cash of the manufacturer is just right equal to recycling and production, and the quantity of carbon quota pledged are just right equal to the carbon allowance (strategy RIIS2):

$$q_{2n}^{(2)} = \frac{n_2\omega_1 - (2kn_2 - c_m{}^2 e_n{}^2 \tau)q_{1n}\sigma}{\omega_3} \tag{A.31}$$

$$\tau_2^{(2)} = \frac{c_m e_n[\omega_1 + \omega_4]}{\omega_3} \tag{A.32}$$

$q_{2n}^{(2)} > 0, q_{2r}^{(2)} > 0$ and $0 < \tau_2^{(2)} < 1$ should be satisfied, when: $\tau > \tau_5$, $\tau > \tau_6$ and $\tau < \tau_7$

① If $\tau_5 - \tau_6 > 0$, thus $c_{m7} < c_m < c_{m6}$, $\tau_5 < \tau < 1$ and $\tau < \tau_7$

② If $\tau_5 - \tau_6 < 0$, thus $c_m < c_{m7}$, $\tau_6 < \tau < 1$ and $\tau < \tau_7$

In this case, we can derive the optimal solutions of $q_{2n}^{(2)b_2}, q_{2r}^{(2)b_2}$ and $\tau_2^{(2)b_2}$ similar with those in strategy RIIA1. That is, strategy RIIS2 is the same as strategy RIIA1 (Table 4).

For R&D investment abatement, substituting the respective optimal production quantities and mitigation rate into the objective function, we obtain the optimal profit of the manufacturer for each case.

A.7. Proof of Theorem 7.

For the carbon purification abatement, according to the optimal production quantities of new and remanufactured products in Theorem 1 and Theorem 2 and the optimal carbon emission reductions and carbon quota pledged in Proposition 1 and Proposition 2, the first derivatives of them in relation to the carbon trading price can be obtained as follows:

In the first period, $\frac{\partial q_{1n}^{(1)}}{\partial c_m}=\frac{1}{2}e_n(\alpha-1)<0$, $\frac{\partial E_{1a}^{(1)}}{\partial c_m}=\frac{1}{2}e_n{}^2\alpha(\alpha-1)<0$

In the second period, ①for the situation of sufficient capital, we select the strategy of better solutions (strategy PIIA2) for comparison, that is : $\frac{\partial q_{2n}^{(1)a_2}}{\partial c_m}=\frac{1}{2}e_n(\alpha-1)<0$, $\frac{\partial q_{2r}^{(1)a_2}}{\partial c_m}=\frac{1}{2}e_n\sigma(\alpha-1)<0$, $\frac{\partial E_{2a}^{(1)a_2}}{\partial c_m}=\frac{1}{2}e_n{}^2\alpha(\alpha-1)(1+\tau\sigma)<0$; ②for the situation of insufficient capital, we select the strategy of better solutions (strategy PIIS3) for comparison, that is : $\frac{\partial q_{2n}^{(1)b_3}}{\partial c_m}=\frac{1}{2}e_n(\alpha-1)<0$, $\frac{\partial q_{2r}^{(1)b_3}}{\partial c_m}=\frac{1}{2}e_n\sigma(\alpha-1)<0$, $\frac{\partial E_{2a}^{(1)b_3}}{\partial c_m}=\frac{1}{2}e_n{}^2\alpha(\alpha-1)<0$

For $\frac{\partial E_f^{(1)b_3}}{\partial c_m}=\frac{1}{2\beta c_m{}^2}[c_pe_n\omega_4+2(\psi_3+c_nk)+\psi_4(\psi_4-2q_{1n}-Q)+2\psi_5]$, $c_pe_n\omega_4>0$, $\psi_4-2q_{1n}-Q<0$, $\psi_5>0$ and ψ_3+c_nk may be positive or negative, thus the monotonicity of $\frac{\partial E_f^{(1)b_3}}{\partial c_m}$ needs to be observed with the graph (see Fig. 2).

A.8. Proof of Theorem 8.

For the R&D investment abatement, according to the optimal production quantities of new and remanufactured products in Theorem 4 and Theorem 5 and the optimal carbon emission reductions and carbon quota pledged in Proposition 3 and Proposition 4, the first derivatives of them in relation to the carbon trading price can be obtained as follows:

In the first period, $q_{1n}^{(2)}=\frac{n_1\omega_1}{\omega_2}$, in which, $\frac{\partial\omega_1}{\partial c_m}=-e_n<0$, $\frac{\partial\omega_2}{\partial c_m}=-2e_n{}^2c_m<0$, we will review the monotonicity of $\frac{\partial q_{1n}^{(2)}}{\partial c_m}$ by the graph. Also, $\tau_1^{(2)}=\frac{c_me_n\omega_1}{\omega_2}$, same as above for the monotonicity of $\frac{\partial\tau_1^{(2)}}{\partial c_m}$. $E_{1a}^{(2)}=\frac{c_me_n{}^2n_1\omega_1{}^2}{\omega_2{}^2}$, in which, $\frac{\partial c_me_n{}^2n_1}{\partial c_m}=e_n{}^2n_1>0$, thus $\frac{\partial E_{1a}^{(2)}}{\partial c_m}>0$.

In the second period, ①for the situation of sufficient capital, $q_{2n}^{(2)a}=\frac{n_2\omega_1-(2kn_2-c_m{}^2e_n{}^2\tau)q_{1n}\sigma}{\omega_2}$, in which, $\frac{\partial\omega_3}{\partial c_m}=-2e_n{}^2c_m<0$, $\frac{\partial(2kn_2-c_m{}^2e_n{}^2\tau)}{\partial c_m}=-2\tau e_n{}^2c_m<0$, the monotonicity of $\frac{\partial q_{2n}^{(2)a}}{\partial c_m}$ should be reviewed by graphs, and the monotonicity of $\frac{\partial q_{2r}^{(2)a}}{\partial c_m}$ is the same as $\frac{\partial q_{1n}^{(2)}}{\partial c_m}$. Also, for $\tau_2^{(2)a}=\frac{c_me_n(\omega_1-\omega_4)}{\omega_3}$, $\frac{\partial c_me_n(\omega_1-\omega_4)}{\partial c_m}=e_n(\omega_1-\omega_4)-c_me_n{}^2$ may be positive or negative, the monotonicity nor sure. $E_{2a}^{(2)a}=\frac{c_me_n{}^2n_2(\omega_1-\omega_4)^2}{\omega_3{}^2}$, in which, $\frac{\partial c_me_n{}^2n_2}{\partial c_m}=e_n{}^2n_2>0$, thus $\frac{\partial E_{2a}^{(2)a}}{\partial c_m}>0$; ②for the situation of insufficient capital, we select the strategy of better solutions (strategy RIIS2) for

comparison. Recall that strategy RIIS2 is the same as strategy RIIA1, which have been proved above. Thus, $q_{2n}^{(2)b_3} = q_{2n}^{(2)a}$, $q_{2r}^{(2)b_3} = q_{2r}^{(2)a}$, $\tau_2^{(2)b_3} = \tau_2^{(2)a}$, $E_{2a}^{(2)b_3} = E_{2a}^{(2)a}$, the monotonicity is no longer stated.

$E_f^{(2)b_3} = \frac{1}{c_m\beta}\{\frac{c_m{}^2 e_n{}^2 n_2(\omega_1-\omega_4)^2}{2\omega_3} + \frac{[n_2\omega_1-(2kn_2-c_m{}^2e_n{}^2\tau)q_{1n}\sigma]c_n}{\omega_3} - \psi_3 + q_{1n}\psi_4\}$, in which, $\frac{\partial\omega_3}{\partial c_m} < 0$, $\frac{\partial(\omega_1-\omega_4)}{\partial c_m} < 0$, $\frac{\partial[n_2\omega_1-(2kn_2-c_m{}^2e_n{}^2\tau)q_{1n}\sigma]}{\partial c_m} = -2c_m e_n{}^2\tau q_{1n}\sigma < 0$, thus the monotonicity unknown.

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